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PERSONAL SKILLS

- **Strong coding skills** in **Java, Python, HTML, CSS, and JavaScript**. I can complete **full-stack development** independently, including frontend interfaces, backend services, and system integration.
- Familiar with the design and implementation of **distributed systems**. I have hands-on experience using MySQL and Redis in distributed scenarios. I am good at **system architecture design, caching strategies, and deployment** in real-world projects.
- Proficient in **Python for data science tasks** such as **data collection, preprocessing, and visualization**. I have experience in **designing, training, and evaluating models**. I understand the basics of deep learning and how common neural network models work.
- Familiar with **Linux** systems. I can **set up environments and deploy projects**, build machine learning pipelines and run model training and debugging, organize and analyze experimental results.
- Able to quickly translate ideas into working prototypes. I have **strong practical skills** for experimentation and implementation in research projects.

EDUCATION

Master | University of Macau

Sep 2025 – Present

Data Science (Artificial Intelligence Applications)

GPA: 3.85/4.0

Major Courses: CISC7026 Introduction to Deep Learning (A), CISC7201 Data Science Programming (A), CISC7021 Applied Natural Language Processing (A-), CISC7204 Data Science and Data Visualization (A-), CISC7013 Principles of Artificial Intelligence (in progress), CISC7202 Practical Machine Learning (in progress), CISC7203 Databases and Data Analysis (in progress), ECEN7107 Data Analysis for Internet of Things (in progress).

Bachelor | Guangdong University of Foreign Studies

2021.09-2025.06

Software Engineering

GPA: 3.84/4.0 (Top 10%)

Major Courses: Design and Analysis of Algorithms (98), Mobile Application Development (98), Network and Information Security (97), Data Mining (97), Java Programming (96), Data Structures (95), Discrete Mathematics (95), Software Engineering (95), Introduction to Cloud Computing and Big Data (94), Principles of Database Systems (93), Linear Algebra (91).

Campus Experience: Served in the Grassroots Development Department of the Youth League Committee from 2021 to 2023, participating in the planning and organization of large-scale university events, with strong communication, collaboration, and execution skills. Volunteered as a teacher at Datang Middle School in Shaoguan, Guangdong, in August 2022.

WORK EXPERIENCE

Baidu YY Live - Software Testing Intern

Dec 2024 – Mar 2025

- Participated in requirement reviews, test case design, functional testing, regression testing, and pre-release smoke testing for modules such as activities and private/group chats, covering the full testing lifecycle.
- Designed boundary and edge test scenarios based on business workflows and existing feature impacts; followed up on defect reporting, reproduction, verification, and regression to ensure release quality.
- Identified and helped locate an emoji encoding issue in the text chat module of the YY Voice app, and facilitated its resolution.
- Completed test design and release support for the “Luoxi New User Tasks Phase II” project; organized product acceptance for the Qingsongjuhe mini program.

University of Macau Centre for Innovation and Entrepreneurship - IT Helper

Jan 2026 - Apr 2026

- Responsible for daily maintenance of the center’s website and the “Big Health+ Accelerator” official website, including content management and page configuration based on WordPress.
- Completed responsive redesign of the official website to support mobile display, enabling the launch of pages for courses, open projects, seminars, and innovation competitions.

PROJECT EXPERIENCE

Dextea Online Order System

Oct 2024 - Apr 2025

Tech Stack: Spring Cloud, Nacos, MySQL, Redis, Vue.js, uni-app, WebSocket, Sa-Token, Alipay SDK, Amap SDK.

Docs: <https://www.kdocs.cn/l/chEBiZP4Sh1X> **Code:** <https://github.com/wilson-lyc/dextea-v2-api>

Responsibility: As my undergraduate graduation project, I independently completed the system design and programming, and implemented the entire project.

Result:

- This project includes **backend services, a webpage** for administrators to manage data, and a **mini-program** for customers to place orders and make payments.
- The system is integrated with the Alipay platform and has **real transaction** capabilities.
- Implemented **nearly store recommendations** using Redis GEO.

- The system is designed with a **multi-tenant architecture** to support chain beverage businesses, enabling **flexible features** such as store-specific configurations and customized product offerings.
- The system implements **RBAC-based access control** at the API level to ensure data and financial security.
- It also uses Redis for **distributed caching** to support high-concurrency access.

Scent4You: Deep Learning-based Perfume Recommendation System

Dec 2025

Tech Stack: Python, PyTorch, Transformers, scikit-learn, pandas, NumPy, Selenium, BeautifulSoup, Plotly, Matplotlib, Jupyter Notebook.

Code: <https://github.com/wilson-lyc/Scent4You>

Result:

- Built a **neural network-based perfume recommendation system** that transforms fragrance notes and perfume structures into continuous vector spaces for semantic retrieval and recommendation.
- **Designed and implemented a complete pipeline** covering perfume data crawling, cleaning, note mapping, embedding generation, model training, evaluation, and recommendation demos.
- **Fine-tuned** bert-base-chinese to build a **Note Embedding Model (NEM)** for Chinese scent-note representation.
- Developed a **Perfume Embedding Model (PEM)** with hierarchical sub-networks, attention aggregation, and multi-head attention to generate perfume embeddings for classification and retrieval.
- Implemented both **weighted-average and neural-network-based recommendation strategies**, using cosine similarity to return Top-10 perfume recommendations based on users' preferred top, middle, and base notes.
- Produced **data analysis and visualization** modules to support market exploration, data understanding, and model interpretation.

HONORS & AWARDS

- Outstanding Graduation Project (Top 5% in College) (2025)
- Outstanding Bachelor of Engineering (2025)
- "Outstanding Communist Youth League Student Cadre" (2022–2023, 2023–2024)
- "Comprehensive Second-Class Scholarship" (2022–2023, 2023–2024)
- First Prize (Backend Track), 17th Quanta Winter Camp (2023)
- "Challenge Cup" Undergraduate Extracurricular Science and Technology Competition, Guangdong University of Foreign Studies (2023)
- Third Prize, National Finals, Team Programming Ladder Competition, China Collegiate Computer Contest (2022)
- "Merit Student" (2021–2022)
- "Outstanding Individual" in "Bringing Culture, Science & Technology, and Health to the Countryside" (2022)